

FULL STACK DOCKER DEPLOYMENT GUIDE

MongoDB + Spring Boot Backend + Angular Frontend

PREREQUISITES

- Docker Desktop (Linux containers)
- Java 17
- Maven
- Node 20+

BACKEND STEPS

1. Build backend

```
mvn clean package -DskipTests
```

2. Dockerfile

```
FROM eclipse-temurin:17-jdk
```

```
WORKDIR /app
```

```
COPY target/*.war app.war
```

```
EXPOSE 8081
```

```
ENTRYPOINT ["java", "-jar", "app.war"]
```

3. Build image

```
docker build -t loan-backend .
```

FRONTEND STEPS

Dockerfile

```
FROM node:20-alpine AS build
```

```
WORKDIR /app
```

```
COPY package*.json ./
```

```
RUN npm install --legacy-peer-deps
```

```
COPY . .
```

```
RUN npm run build
```

```
FROM nginx:alpine
```

COPY --from=build /app/dist/loan-ui/browser /usr/share/nginx/html

EXPOSE 80

RUN COMMANDS

docker run -d --name mongo -p 27017:27017 mongo:7

docker run -d --name loan-backend-container -p 8081:8081 --link mongo:mongo loan-backend

docker run -d --name loan-frontend-container -p 4200:80 loan-frontend

VERIFY

Frontend: <http://localhost:4200>

Backend: <http://localhost:8081>

STOP

docker stop mongo loan-backend-container loan-frontend-container